

Laboratory Notebook Rubric

PHYS 201L | Fall 2026 | One notebook submitted per team

Team / authors:	Experiment:	Evaluator:	Final score: ____ / 100
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Scoring: Select a performance level for each criterion. Award up to the listed weight; the descriptors define the evidence expected. A notebook that does not run from top to bottom, omits requested raw data, or lacks required team identification is incomplete until corrected.

Criterion	Wt.	0–1: Missing / insufficient	2: Developing	3: Proficient	4: Excellent
Purpose, model, and prediction Score: ____ / 10	10	Purpose is missing or copied without evidence of understanding. Relevant physical model, variables, or prediction is absent or incorrect.	Purpose and model are present but incomplete. Prediction or expected relationship is weakly justified or disconnected from the analysis.	Purpose, physical model, variables, and expected relationship are correct and clearly connected to the experiment.	Provides a concise, insightful rationale; develops the correct model and a quantitatively testable prediction that guides the analysis.
Experimental methods and data integrity Score: ____ / 15	15	Procedure is missing, inaccurate, written only as generic instructions, or cannot explain how the team obtained its data. Required raw data are missing.	Describes most actions but omits important apparatus settings, calibrations, repeated trials, or changes made during data collection.	Accurately describes what the team did. Apparatus, calibration, measurements, repeated trials, and deviations are sufficient for reproduction.	Provides a reproducible account and explains quality-control decisions, problems encountered, diagnostic checks, and effective corrections.
Calculations and computational analysis Score: ____ / 20	20	Required calculations are absent or substantially incorrect. Code fails, uses inappropriate formulas, or produces results that were not checked.	Core calculations are attempted but contain errors, unclear units, hard-coded inconsistencies, or analysis steps that are difficult to follow.	Calculations and code are correct, organized, unit-consistent, and traceable from raw data to reported results. Requested sample calculations are included.	Analysis is correct and efficient; code is readable and reusable, intermediate checks are meaningful, and results are independently sanity-checked.

Laboratory Notebook Rubric

PHYS 201L | Criteria continued

Criterion	Wt.	0–1: Missing / insufficient	2: Developing	3: Proficient	4: Excellent
Uncertainty analysis and quantitative comparison Score: ____ / 20	20	Uncertainties are missing, physically unjustified, or propagated incorrectly. Theory and experiment are compared only subjectively.	Some uncertainties and numerical comparisons are present, but important sources, propagation steps, error bars, or interpretations are incomplete.	Measurement uncertainties are justified and propagated correctly. Experimental and predicted/accepted results are compared quantitatively.	Uncertainty treatment is complete and physically insightful; comparisons use appropriate error bars, percent differences, fit parameters, or consistency tests.
Tables, figures, and presentation of evidence Score: ____ / 15	15	Required tables or figures are missing or misleading. Axes, units, uncertainties, captions, or legends are absent; screenshots replace reproducible output.	Evidence is present but formatting, labels, units, captions, scaling, or error bars are inconsistent or difficult to interpret.	Tables and figures are readable and correctly labeled, with units, uncertainties/error bars, appropriate scales, captions, and fit information where required.	Visual evidence is exceptionally clear and economical; design choices reveal the physical relationship and make comparisons immediately understandable.

Laboratory Notebook Rubric

PHYS 201L | Criteria and feedback

Criterion	Wt.	0–1: Missing / insufficient	2: Developing	3: Proficient	4: Excellent
Interpretation and conclusions Score: ____ / 15	15	Conclusion is missing, merely repeats results, makes unsupported claims, or fails to answer required questions. Uses vague 'human error' explanations.	States a plausible conclusion and answers most questions, but connections among evidence, model, uncertainty, and limitations are incomplete.	Uses numerical evidence to support conclusions, answers all required questions, explains limitations, and identifies realistic improvements.	Builds a compelling evidence-based argument, distinguishes random and systematic limitations, and proposes specific improvements or follow-up tests.
Scientific communication and reproducibility Score: ____ / 5	5	Team/date information is missing; prose is difficult to follow; notebook contains broken cells, stale outputs, undocumented outside assistance, or private information.	Notebook is understandable but has organizational, grammar, formatting, citation, or execution-order problems.	Notebook is clear, concise, professional, properly attributed, and runs in order with final outputs saved. No sensitive information is included.	Notebook is polished and exceptionally easy to audit, rerun, and learn from; prose, equations, code, and evidence form one coherent scientific narrative.

Required completion checks

☐ All team members and dates identified	☐ Notebook runs top to bottom	☐ Final outputs and graph(s) saved	☐ Raw data / appendix included	☐ Material assistance acknowledged
Evaluator note:	Evaluator note:	Evaluator note:	Evaluator note:	Evaluator note:

Overall feedback: _____